

KARNATAKA RESIDENTIAL EDUCATIONAL INSTITUTIONS SOCIETY

ICT Class 7 - Formative Assessment 3

Formative Assessment (FA-3) - Answer Key

Class: Class 7

Assessment Type: Formative Assessment (FA-3)

Total Marks: 30

Duration: 90 minutes

Topics Covered: Programming - Scratch, Digital Citizenship

Answer Key - Grading Guidelines

Note to Teacher: This answer key provides expected answers and marking guidelines. Accept equivalent answers where appropriate. Partial marks may be awarded for incomplete but correct responses.

Part A: Written Concept Paper (20 Marks)

Section I: Multiple Choice Questions (5 marks)

Q.No	Answer	Explanation
1	a) create clone of myself	This block creates a duplicate copy of the sprite at runtime.
2	b) The trail of data left by online activities	A digital footprint is the record of all online activities a user engages in.
3	b) Hacking into someone's account	Hacking is unauthorized access to someone's computer or account, which is a cybercrime.
4	b) Runs the code repeatedly until stopped	The forever block continuously executes the blocks inside it until the program is stopped.
5	b) To protect the account from unauthorized access	Strong passwords prevent unauthorized users from accessing accounts.

Section II: Short Answer Questions (10 marks)

1. What are “clones” in Scratch? How are they created?

- **Answer:** Clones are duplicate copies of a sprite created during runtime. They are created using the “create clone of myself” block. Clones can have their own scripts and behave independently from the original sprite.
- **Marking:** 1 mark for definition, 1 mark for creation method.

2. Name any two rules of cyber safety that every student should follow.

- **Answer:** 1) Never share personal information (name, address, phone number) online. 2) Never open emails or links from unknown senders. 3) Use strong passwords and change them regularly. 4) Always log out from accounts after use.
- **Marking:** 0.5 marks for each correct rule (1 mark for two rules).

3. What is the difference between a “local variable” and a “global variable” in Scratch?

- **Answer:** A local variable is accessible only within the sprite it is created for. A global variable (For all sprites) can be accessed and modified by any sprite in the project. Local variables are useful for sprite-specific data, while global variables are used for shared data like scores.
- **Marking:** 1 mark for each variable type explanation.

4. What is phishing? How can you identify a phishing email?

- **Answer:** Phishing is a fraudulent attempt to obtain sensitive information like passwords and credit card details by pretending to be a trustworthy entity. Signs of phishing emails: urgent language, misspelled URLs, requests for personal information, unknown sender addresses, and suspicious attachments.
- **Marking:** 1 mark for definition, 1 mark for identification signs.

5. **Why is it important to create a custom block in Scratch? Give one advantage.**

- **Answer:** Custom blocks allow you to create reusable code that can be called multiple times. One advantage is that it makes the code more organized and easier to debug. You can also pass parameters to make the block flexible for different situations.
- **Marking:** 1 mark for importance, 1 mark for advantage.

Section III: Long Answer Questions (5 marks)

1. **Describe the steps to create a simple “Catch the Falling Objects” game in Scratch. Explain the role of at least four different blocks used in the game.**

- **Answer:**
 - ▶ Step 1: Create a cat sprite as the catcher and a star sprite as the falling object.
 - ▶ Step 2: Use “when green flag clicked” event block to start the game.
 - ▶ Step 3: Use “forever” loop to make the star fall continuously.
 - ▶ Step 4: Use “change y by -5” block inside the loop to make the star move downward.
 - ▶ Step 5: Use “if key right arrow pressed” and “change x by 10” blocks to move the cat.
 - ▶ Step 6: Use “touching” block to detect collision and increase score.
 - ▶ Step 7: Use “if y position < -170” to detect when star reaches bottom and show “Game Over”.
- **Marking:** 1 mark for each step explained, 2 marks for explaining four blocks.

Part B: Hands-on Practical Lab Task (5 marks)

1. **Create a Scratch project where a cat sprite chases a mouse sprite using arrow keys. Use the “point towards” and “move” blocks.**
 - **Expected:** Cat sprite follows mouse, uses “point towards mouse” and “move steps” blocks, arrow key controls work.
 - **Marking:** 1 mark for sprite setup, 1 mark for correct blocks used.
2. **Create a Scratch project that uses at least 3 clones to display multiple stars on the screen. Use the “create clone of” block.**
 - **Expected:** At least 3 star clones visible on screen, “create clone of” block used correctly.
 - **Marking:** 1 mark for clone creation, 1 mark for correct implementation.
3. **Save your Scratch projects on the desktop with the filenames “ChaseGame.sb3” and “ClonesDemo.sb3”.**
 - **Expected:** Both files saved on desktop with correct filenames.
 - **Marking:** 0.5 marks for each correct file.

Part C: Notebook Maintenance (5 marks)

Marks will be awarded for:

- Regularity in completing classwork (2 marks)
- Neatness of notes and diagrams (2 marks)
- Completion of all class activities (1 mark)